

Factors Influencing the Median Home Value Across U.S. Metropolitan Areas

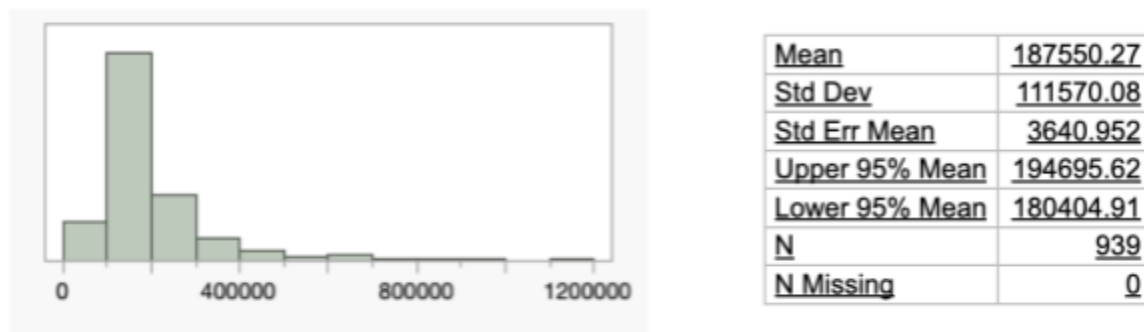
ILRST 2130 Project Component 2

Nichole Zheng (nz268)

Before conducting any formal modeling, this component provides an exploratory analysis of the variables used in the study. The dataset contains observations for approximately 940 U.S. metropolitan and micropolitan areas, with variables capturing economic conditions, demographics, and housing market structure. This exploratory analysis summarizes the distributions of each variable and examines pairwise relationships to better understand patterns in the data prior to regression modeling.

Distribution of the Response Variable

Figure 1. Histogram & Summary Statistics of Median Home Values

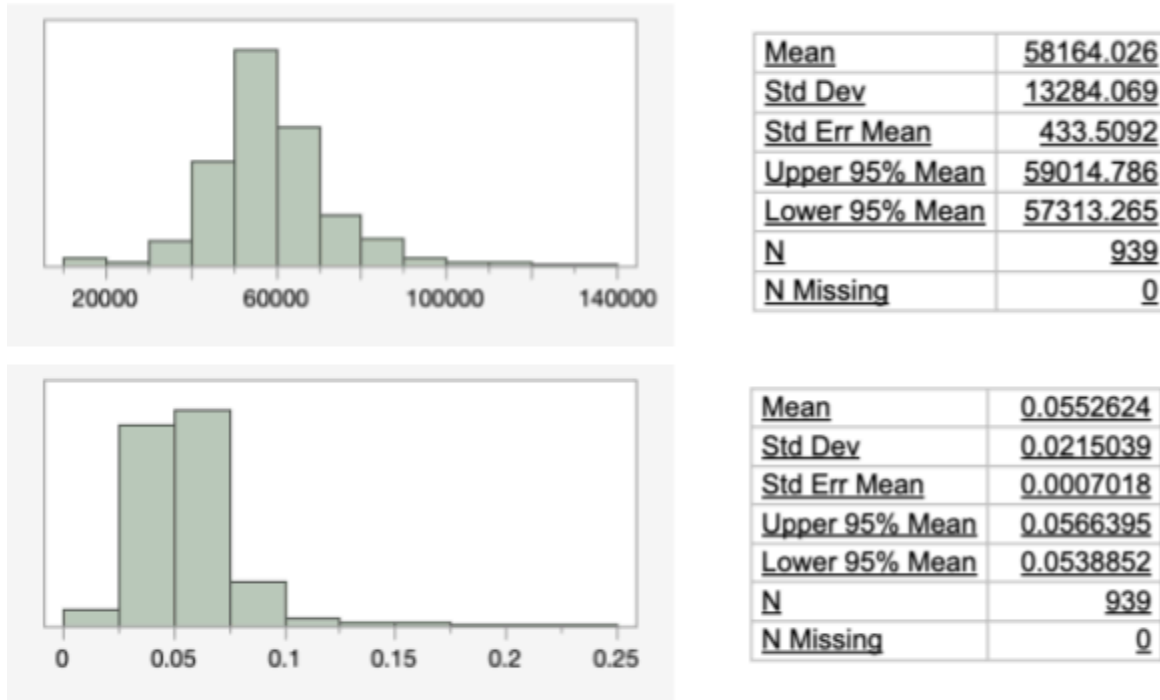


The histogram in figure 1 displays the distribution of median home values across U.S. metropolitan and micropolitan areas. The distribution is right-skewed, with a long upper tail corresponding to a relatively small number of metropolitan areas with very high home values. The median home value is approximately \$156,300, while the mean is higher at about \$187,550, indicating that high-value outliers pull the mean upward.

The interquartile range spans from approximately \$122,000 (25th percentile) to \$213,900 (75th percentile), showing substantial variation in typical home values across metropolitan areas. The minimum median home value is about \$62,000, while the maximum exceeds \$1.1 million, further highlighting the presence of extreme high-value markets. Overall, this distribution suggests considerable variability in housing prices across metropolitan areas, with most areas clustered at lower values and a small number of high-cost markets contributing to the upper tail.

Distribution of the Predictor Variable

Figure 2. Distributions of economic and labor market predictors

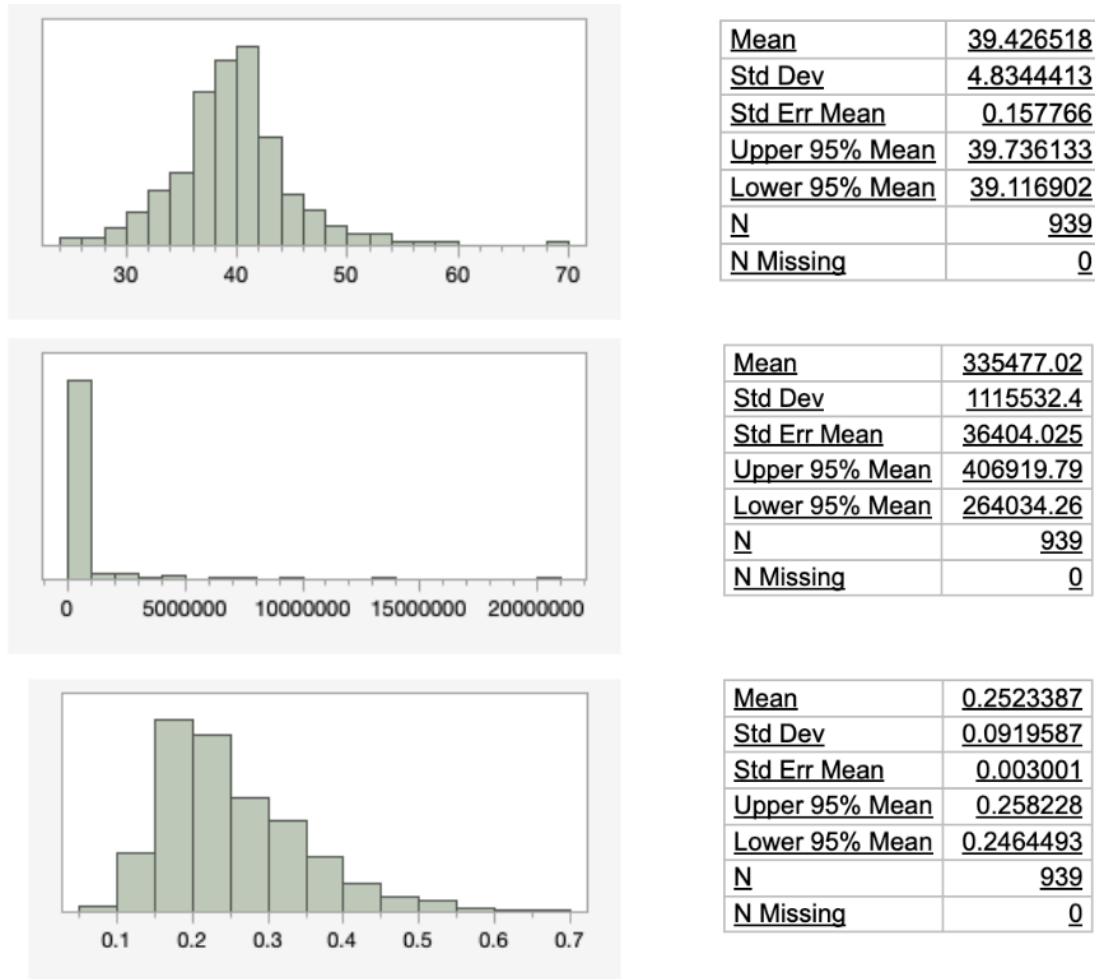


Note. The top panel shows the distribution of median household income, and the bottom panel shows the distribution of the unemployment rate.

The top panel of Figure 2 shows the distribution of median household income, which is slightly right-skewed, with most metropolitan areas clustered between approximately \$50,000 and \$65,000. The median household income is about \$57,055, and the mean is slightly higher at \$58,164, indicating some skewness driven by higher-income metropolitan areas. While most areas fall within a relatively narrow income range, a small number of high-income regions extend the upper tail of the distribution.

The bottom panel of Figure 2 displays the distribution of the unemployment rate, which is also right-skewed. Most metropolitan areas have unemployment rates between approximately 4% and 7%, with a mean unemployment rate of about 5.5%. However, a smaller number of areas have substantially higher unemployment rates, creating a long right tail that extends beyond 10%. This variation suggests key differences in labor market conditions across metropolitan areas, which may influence housing demand and home values.

Figure 3. Distributions of demographic predictors



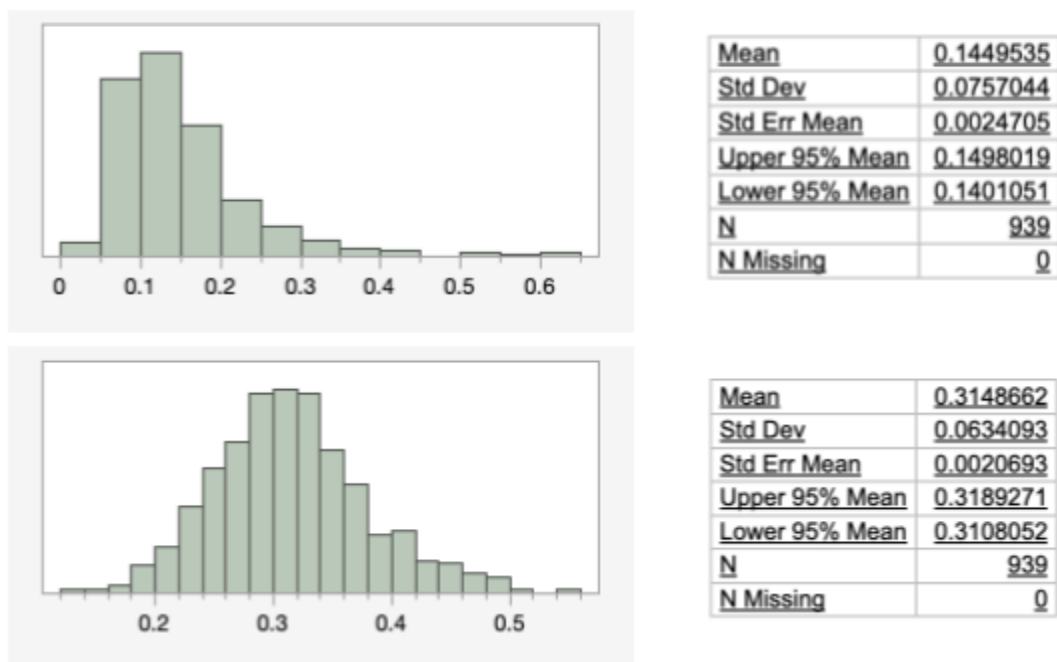
Note. The top panel shows the distribution of total population, the middle panel shows the distribution of total population and the bottom panel shows the distribution of the unemployment rate.

Figure 3 summarizes the distributions of median age, total population, and the percentage of the population with a bachelor's degree or higher across U.S. metropolitan and micropolitan areas. The distribution of median age is approximately symmetric, centered around a mean of 39.4 years, with most metropolitan areas falling between roughly 37 and 42 years. This relatively narrow spread indicates that age composition varies modestly across metropolitan areas compared to other demographic characteristics.

The distribution of total population is highly right-skewed, with a median population of approximately 75,139 and a mean of 335,477, reflecting the presence of a small number of very large metropolitan areas. While most metropolitan areas have relatively small populations, a few large urban regions substantially extend the upper tail of the distribution, with the largest exceeding 20 million residents.

The distribution of the percentage of residents with a bachelor's degree or higher is right-skewed, with a median value of approximately 23.1% and a mean of 25.2%. Most metropolitan areas have between 18% and 30% of adults holding at least a bachelor's degree, while a smaller number of highly educated areas exhibit substantially higher levels of educational attainment. Overall, these distributions highlight meaningful demographic variation across metropolitan areas that may be associated with differences in housing market outcomes.

Figure 4. Distributions of Housing Market Structure Variables



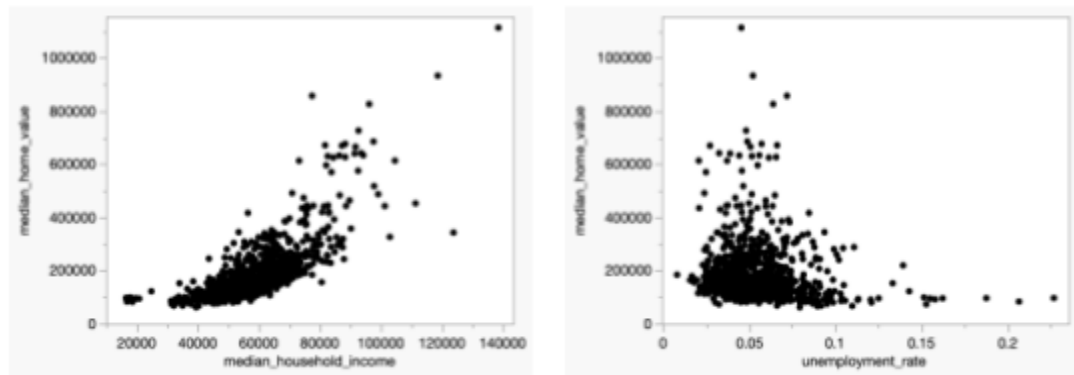
Note. The top panel shows the distribution of median household income, and the bottom panel shows the distribution of the unemployment rate.

The vacancy rate distribution is strongly right-skewed, with most metropolitan areas clustered at low vacancy levels and a long tail representing a smaller number of areas with substantially higher vacancy rates. The median vacancy rate is approximately 12.9%, while the mean is 14.5%. Although the majority of areas have vacancy rates below 20%, a long right tail extends beyond 40%, indicating a significant difference in housing supply conditions across regions.

The distribution of the percentage of renter-occupied housing units is more symmetrical, centered around a median of approximately 31.1% and a mean of 31.5%. Most metropolitan areas fall between roughly 27% and 35% renter-occupied housing, with fewer areas at the extremes. This suggests that while the share of renters varies across metropolitan areas, it does so within a more limited range compared to vacancy rates. Together, these distributions highlight meaningful differences in housing market structure that may be closely related to variation in median home values across metropolitan areas.

Bivariate Relationships

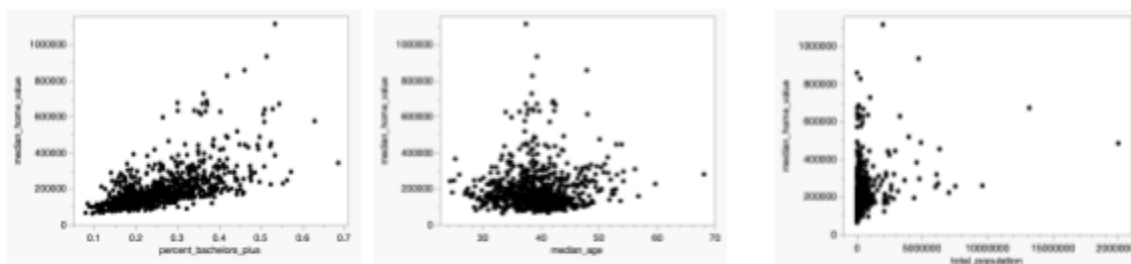
Figure 5. Median home value versus economic predictors



Note. The right panel displays median home value versus median household income, and the left panel displays median home value versus the unemployment rate.

Figure 5 shows the relationship between median home value and two economic indicators: median household income and the unemployment rate. Median home value shows a strong positive association with median household income, with higher-income metropolitan areas generally corresponding to substantially higher home values. The relationship appears approximately linear, even though dispersion increases at higher income levels. This suggests that income alone does not fully explain variation in housing prices. In contrast, the relationship between median home value and unemployment rate is negative but weaker, with higher unemployment rates generally associated with lower home values. Considerable variability remains at lower unemployment levels, indicating that labor market conditions interact with other factors in influencing housing prices.

Figure 6. Median home value versus demographic predictors



Note. The right panel displays median home value versus percent of people with bachelor's degree or higher, the middle panel displays the median home value versus median age, and the left panel displays median home value versus the total population.

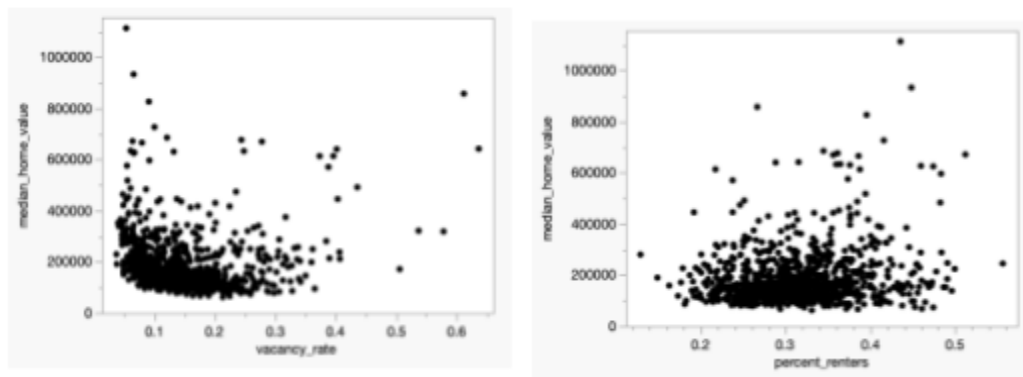
Figure 6 presents scatterplots of median home value against demographic characteristics, including educational attainment, median age, and total population. Median home value shows a clear positive association with the percentage of residents holding a bachelor's degree or higher, with more highly educated metropolitan areas generally

corresponding to higher housing prices. The relationship appears moderately strong, though variability increases at higher levels of educational attainment.

In contrast, the relationship between median home value and median age is weaker and more dispersed. While some metropolitan areas with older populations exhibit higher home values, the overall pattern is not strongly linear, suggesting that age composition alone does not explain much of the variation in housing prices.

The scatterplot of total population versus median home value shows a highly scattered pattern with no clear linear relationship. Most metropolitan areas have relatively small populations, with a few very large urban areas forming a long right tail. Although some large metropolitan areas have high home values, the relationship is inconsistent, indicating that population size by itself is not a strong predictor of housing prices across metropolitan areas.

Figure 7. Median home value versus housing market structure



Note. The right panel displays median home value versus vacancy rate, and the left panel displays median home value versus the percent renters.

The figure above displays the relationship between median home value and housing market structure variables, including vacancy rate and the percentage of renter-occupied units. The scatterplot of vacancy rate versus median home value shows a generally negative association, where metropolitan areas with higher vacancy rates tend to have lower home values. The relationship is somewhat dispersed, but areas with very high vacancy rates rarely exhibit high median home values, suggesting that weaker housing demand or excess supply may be linked to lower prices.

The scatterplot of percent renters versus median home value shows a weak and less consistent relationship. While some areas with higher renter shares have lower home values, there is substantial variation across the full range of renter percentages. Metropolitan areas with similar renter proportions can have very different housing prices, indicating that renter share alone does not strongly predict median home value.

Overall, these results suggest that vacancy rate may be more closely related to housing prices than renter share, although both variables exhibit considerable variability across metropolitan areas.

Correlation Analysis

	median_home_value	percent_renters	median_household_income	median_age	unemployment_rate	percent_bachelors_plus	vacancy_rate	total_population
median_home_value	1.0000	0.1879	0.7539	0.0334	-0.1572	0.6345	-0.0060	0.3148
percent_renters	0.1879	1.0000	-0.0172	-0.6149	0.1879	0.2588	-0.2246	0.2239
median_household_income	0.7539	-0.0172	1.0000	-0.0298	-0.4730	0.6353	-0.2672	0.3183
median_age	0.0334	-0.6149	-0.0298	1.0000	0.0557	-0.1304	0.3303	-0.0654
unemployment_rate	-0.1572	0.1879	-0.4730	0.0557	1.0000	-0.2652	0.2615	0.0154
percent_bachelors_plus	0.6345	0.2588	0.6353	-0.1304	-0.2652	1.0000	-0.0902	0.3004
vacancy_rate	-0.0060	-0.2246	-0.2672	0.3303	0.2615	-0.0902	1.0000	-0.1793
total_population	0.3148	0.2239	0.3183	-0.0654	0.0154	0.3004	-0.1793	1.0000

The correlation matrix above reports Pearson correlation coefficients among median home value and the selected economic, demographic, and housing market predictors. Median home value exhibits a strong positive correlation with median household income ($r = 0.754$) and with the percentage of residents holding a bachelor's degree or higher ($r = 0.635$), indicating that metropolitan areas with higher incomes and higher levels of educational attainment tend to have higher housing prices. Median home value also shows a moderate positive correlation with total population ($r = 0.315$), suggesting that larger metropolitan areas may generally experience higher home values.

Median home value is weakly negatively correlated with the unemployment rate ($r = -0.157$), implying that higher unemployment is somewhat associated with lower home values, though the relationship is modest. The correlation between median home value and vacancy rate is near zero ($r = -0.006$), indicating little linear association between these variables despite the negative pattern observed in the scatterplot. This suggests that vacancy rate may have a more complex or indirect relationship with housing prices.

Among the predictors, median household income and educational attainment are strongly positively correlated ($r = 0.635$), indicating that areas with more educated populations also tend to have higher incomes. Median household income is moderately negatively correlated with unemployment rate ($r = -0.473$), which is consistent with general labor market dynamics. These relationships suggest some potential for multicollinearity among economic predictors, which should be considered in regression modeling.

Based on the exploratory analysis and correlation results in Component 2, the regression model will most likely use median home value as the response variable and include median household income, percentage of residents with a bachelor's degree or higher, and vacancy rate as predictor variables. These predictors were selected because they exhibit strong relationships with median home value and represent distinct economic and housing market mechanisms. Other variables were excluded due to weaker associations with the response variable or potential redundancy with the selected predictors. This specification provides a clear and interpretable model for explaining variation in median home values across U.S. metropolitan areas.